

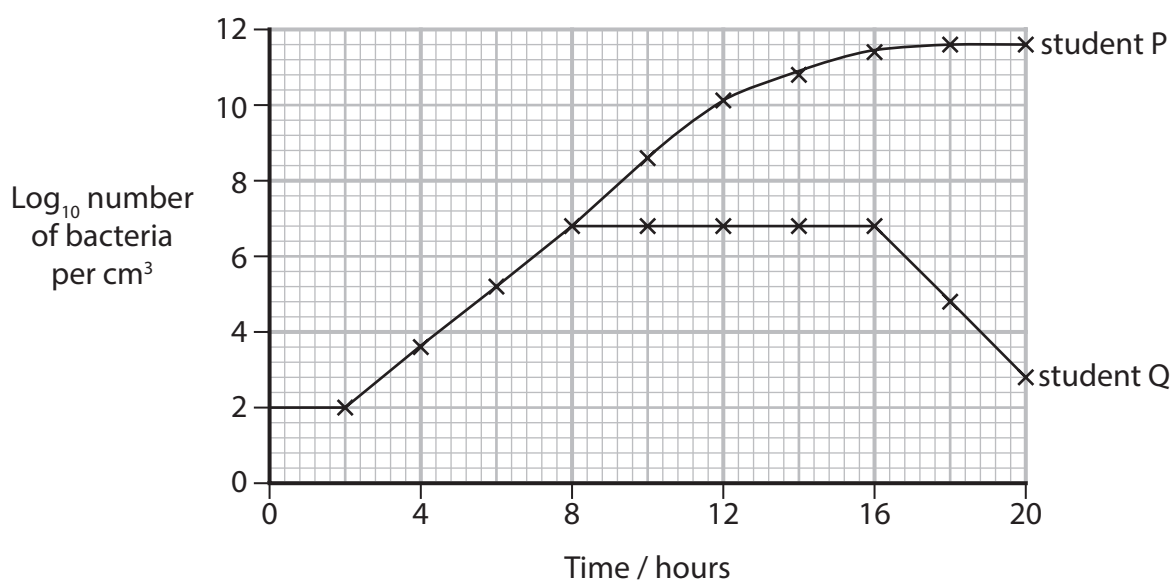
- 2 Two students, student P and student Q, each cultured bacteria in a liquid medium for 20 hours.

The culture conditions used by both students were identical.

Every two hours, each student removed a sample of the culture to count the number of bacteria.

Each student used a different method to count the number of bacteria in the samples.

The graph shows the bacterial growth curve that each student drew using their results.



- (a) Both students used aseptic techniques to culture the bacteria.

- (i) Explain the importance of using aseptic techniques to culture bacteria.

(2)

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(ii) Explain **one** aseptic technique that should be used when culturing bacteria.

(2)

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(b) Calculate the growth rate constant (k) of these bacteria between 2 hours and 6 hours of culture.

(2)

Use the equation: $k = \frac{\log_{10} N_t - \log_{10} N_0}{0.301 \times t}$

where:

$\log_{10} N_t = \log_{10}$ number of cells at 6 hours

$\log_{10} N_0 = \log_{10}$ number of cells at 2 hours.

Answer

(c) The number of bacteria in the samples taken after 10 hours were:

398 107 171 in the culture of student P

6 309 573 in the culture of student Q.

(i) Calculate the ratio of the number of bacteria in these two cultures.

(1)

Answer : 1

- (ii) Student P and student Q used different methods for determining the number of bacteria.

State which method each student used.

Give a reason for your answer.

(2)

Student P

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Student Q

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(Total for Question 2 = 9 marks)